

Pathmika Weeraratna

Associate Tech Lead — AI/ML Engineer

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Summary

AI/ML engineer and Associate Tech Lead with nearly five years of software engineering experience, having self-initiated and led applied AI/ML capabilities in telecom fraud detection from the ground up at Omobio, progressing from individual learning to leading an AI/ML R&D team. Contributed to 25+ client and product engagements – delivery, architecture, and pre-sales solutioning – across 9+ countries in Asia, Africa, and Latin America. Hands-on experience building production NLP/embedding-based detection systems, GenAI subsystems on local LLMs, and ensemble ML classifiers, backed by strong software engineering fundamentals across full-stack and backend development. Currently pursuing an MSc in Artificial Intelligence at the University of Moratuwa. Recognized as Emerging Employee of the Year (2023).

Experience

Associate Tech Lead

Jan 2026 – Present

Omobio (Pvt) Ltd

- Lead the AI/ML R&D team enhancing an existing rule-based SMS Firewall product with machine-learning-driven detection capabilities, mentoring a cross-functional team of 10+ developers.
- Serve as technical subject-matter expert for AI/ML pre-sales engagements, producing architecture proposals and technical validation for telecom fraud-detection and predictive-analytics platforms across Africa, South Asia, and the Middle East.
- Drive solution design and delivery of scalable AI/ML applications, debugging, testing, and refining models and services through CI/CD-driven workflows.
- Also lead the SOR (Steering of Roaming) centralized reporting platform, a full-stack telecom data-reporting solution spanning backend pipelines and web dashboards.

Senior Software Engineer

Jan 2024 – Jan 2026

Omobio (Pvt) Ltd

- Self-initiated and led AI/ML capabilities in telecom security from the ground up, progressing from learning to team leadership; built a production SMS fraud-detection module (live deployment, Orange/Moov Côte d'Ivoire) combining SentenceTransformer embeddings (`all-MiniLM-L6-v2`) and cosine-similarity matching against a 365-day rolling reference table to auto-block OTP-bypass and grey-route A2P traffic, sustaining ~378 messages/sec.
- Extended the fraud-detection engine with a local-LLM GenAI subsystem (Ollama-hosted Llama 3 8B / Google Gemma via LangChain-Ollama) generating new unallowed-content variants for content review and translation, backed by multilingual BAAI/bge-m3 embeddings for semantic similarity search, and an NLLB-200/FastText pipeline for 176-language identification and translation.
- Built an ensemble ML classifier combining XGBoost and CatBoost with SHAP explainability and Optuna hyperparameter tuning for fraud/anomaly-detection use cases; authored a TPS-based production hardware-sizing model (2–1,200+ TPS scaling tables, GPU-tier flagging) and a tiered data-retention redesign cutting steady-state database growth by ~90%.
- Contributed AI/ML risk-scoring components (Random Forest classification, social-graph-based SIM-box-ring detection via community/centrality analysis) to an R&D flash-call fraud-detection system built on CAMEL/CAP signaling.
- Built a working churn/CLV-prediction prototype (scikit-learn Random Forest, 80–90% validation accuracy) with a Flask API and React dashboard, supporting a 16-module predictive-analytics platform proposal for a telecom operator (Nepal).
- Mentored a team of 10 junior developers on ML engineering practices, code review, and CI/CD (GitHub Actions).

Software Engineer

Apr 2022 – Jan 2024

Omobio (Pvt) Ltd

- Led design and development of cross-platform mobile applications (Hutch Selfcare, Vodafone Selfcare, MyDialog, RetailHub, Circadian, SFA) in React Native, managing releases across Play Store, App Store, and Huawei AppGallery.
- Built responsive web applications with REST API integrations (Next.js, React.js) for enterprise and telecom clients, establishing the full-stack foundation later applied to AI/ML system development.
- Received the Emerging Employee of the Year Award (2023) for outstanding technical contribution.

- Supported development of frontend, backend, and mobile features for enterprise-grade projects using React.js, React Native, Python, and RESTful APIs; transitioned to a full-time Software Engineer role based on strong performance.

Key Projects

- **SMS Firewall AI/ML Suite, 2025–Present:** Modular FastAPI-based detection suite for OTP-bypass, phishing/Unicode analysis, and grey-route identification, processing high-volume SMS data streams; combines semantic embeddings, a local-LLM GenAI layer, and an XGBoost/CatBoost ensemble with SHAP/Optuna.
- **Flash Call / Voice Firewall ML Layer, 2025:** Random Forest risk scoring and social-graph-based SIM-box-ring detection layered on a CAMEL/CAP signaling fraud-detection system; scoped a next-phase Voice Firewall suite (IRSF, Wangiri, SIM-swap).
- **Predictive CLV & Campaign Insights, 2025:** Designed a 16-module ML platform architecture (predictive CLV, churn risk, offer recommendation, uplift modeling, behavioral clustering) and built a working Random Forest churn-prediction prototype with a Flask API and React GUI.

Additional Experience & Engagements

Beyond AI/ML, built production mobile and web applications in React Native and Next.js/React — including a custom native alarm engine, HOPP's payment-integrated web portal, and 2+ years owning Hutch Selfcare (RN version upgrade 0.66.5 → 0.76.5, 3+ feature change requests) — plus personal backend R&D APIs in ASP.NET Core and Spring Boot, full-stack fundamentals underpinning the ML work above.

Also contributed AI/ML solution design and pre-sales architecture to 7+ further engagements: a CVM-BI predictive-analytics platform (multi-region telecom group), a GOV Disaster Management AI/ML alert system, the Grameenphone CDR search platform (Bangladesh), an Ethio Telecom roaming AI/ML suite (Ethiopia), BSNL Selfcare fraud/anomaly-detection AI/ML (India), Bankai Digital Wallet eKYC/credit-scoring AI, and an NWDAF-AF-SOR integration study (Mobileum).

Areas of Expertise

- **AI/ML & Data Science:** PyTorch, Hugging Face Transformers, LangChain, Ollama, XGBoost, CatBoost, scikit-learn, SHAP, Optuna, Random Forest
- **NLP & GenAI:** SentenceTransformers (BAAI/bge-m3), semantic search, FastText, NLLB translation, local-LLM deployment (Llama 3, Google Gemma), RAG (Haystack)
- **Applied ML Domains:** Fraud/anomaly detection, predictive modelling (CLV/churn), social-graph analysis, recommendation systems
- **Backend & APIs:** Python, FastAPI, REST API design, Java (Spring Boot), .NET (ASP.NET Core), Microservices
- **Data Engineering:** PostgreSQL, ClickHouse, MariaDB, Elasticsearch/ELK, Apache Airflow, Apache Spark, Celery
- **Web & Mobile:** React.js, Next.js, TypeScript, React Native
- **DevOps:** Docker, Git, CI/CD (GitHub Actions, Jenkins), Linux

Education

Master of Science in Artificial Intelligence (Reading) – University of Moratuwa July 2025 – Present (Expected 2027)

B.Sc. in Information Systems, Second Class Upper Division, GPA 3.44 – University of Colombo School of Computing
2019 – 2022

Diploma in English (Certificate Level), Distinction Pass – Seagis Campus, Nugegoda

2018

Achievements

- Emerging Employee of the Year – Omobio (2023): recognized for exceptional job performance and contribution to key software engineering projects.
- Director's List Awardee – University of Colombo (2021): achieved GPA above 3.7 in first academic year.
- Authored a 12-lesson internal Telecom AI/ML training curriculum and delivered sessions to engineering teams.

References available upon request.